

NTG-Agent Retrieval Trials

Three live trials against LightRAG's four query modes, with every retrieved document and generated answer kept alongside the numbers they produced.

ENGINE	LightRAG 1.4.16
GENERATOR	gpt-5.1 • azure_openai
EMBEDDINGS	text-embedding-3-large
RERANKER	none configured

SELECT A TRIAL

Validation trial

6 queries × 4 modes · cold cache

HIT-RATE@5 · HYBRID 1.000 target ≥ 0.80	HIT-RATE@1 · HYBRID 0.833 naive reaches 0.833	HYBRID RETRIEVAL COST 4.0× vs naive · cache cold	BACKEND UPTIME 100% 9 samples, 0 outages	MODES SEPARATED No identical top-5 across modes
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Do the modes actually differ?

The question Section 4 asks, and the one the headline number answers badly.

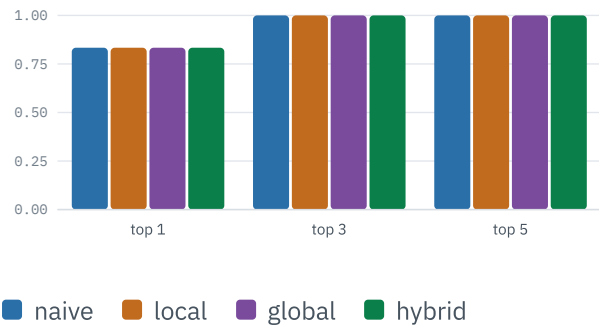
Hit-rate@5 cannot answer this question on this corpus

All four modes returned the **same top-5 documents for every query**, so they score identically at 1.000. The Section 4 criterion “hybrid ≥ best single mode” therefore passes on a tie against naïve, not on evidence that hybrid retrieves better.

Ranking differences do appear at **k = 1**. Below that cutoff the four modes are indistinguishable on this corpus, so the honest comparison between them is cost, not quality.

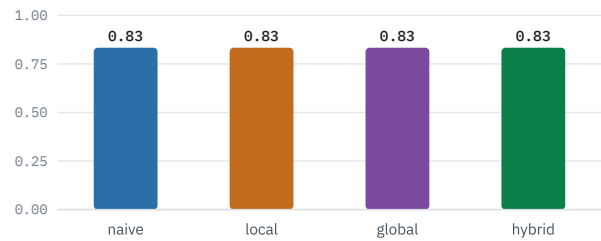
Hit-rate by rank cutoff

Share of queries where a gold document appears in the top k retrieved chunks.



Where ranking does differ

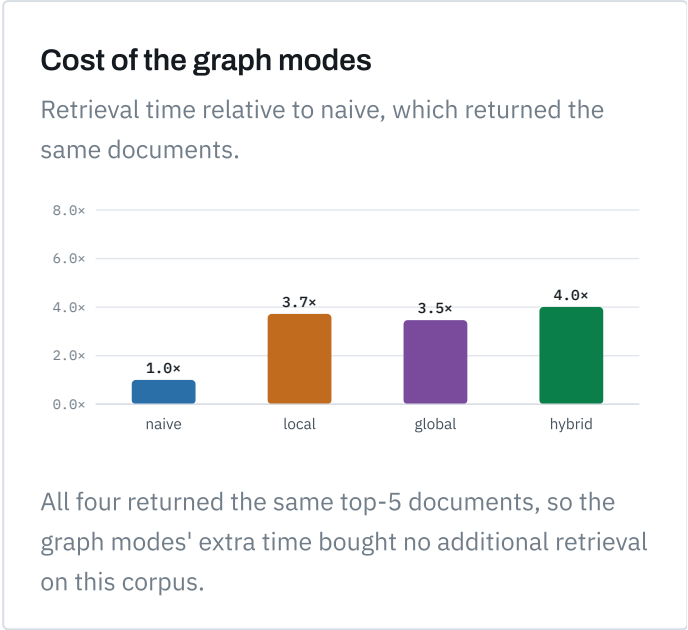
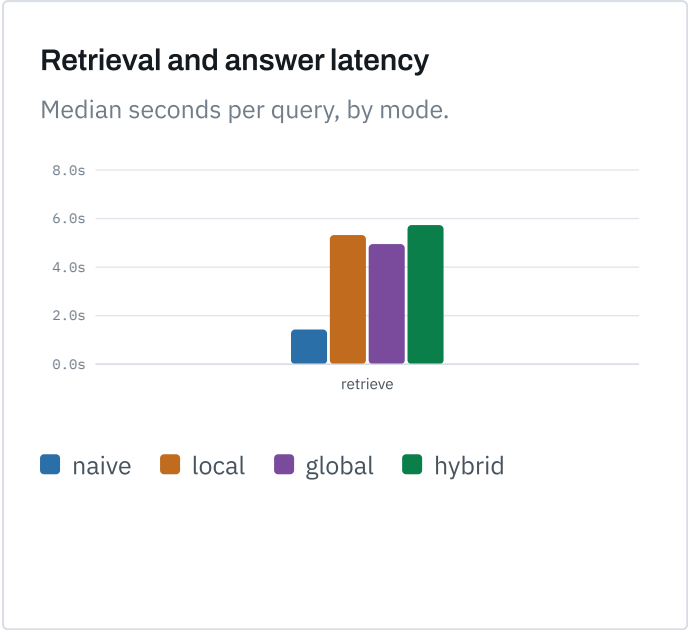
Hit-rate at k = 1, the only cutoff that separates the modes.



By the top-5 cutoff every mode converges on 1.000.

What the modes cost

Identical retrieval, different time to get it — and the cache changes the answer.



HIT-RATE@5 · HYBRID 0.917 target ≥ 0.80	HIT-RATE@1 · HYBRID 0.500 naive reaches 0.583	HYBRID RETRIEVAL COST 3.9× vs naive · cache cold	BACKEND UPTIME 100% 44 samples, 0 outages	MODES SEPARATED No identical top-5 across modes
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Do the modes actually differ?

The question Section 4 asks, and the one the headline number answers badly.

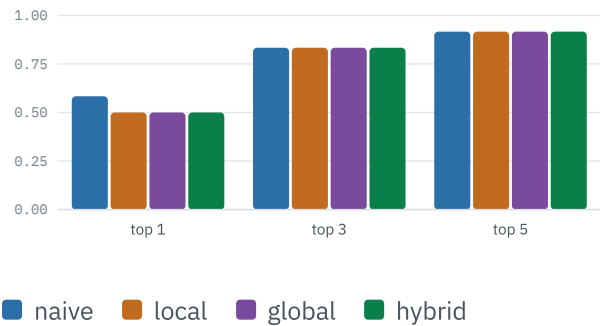
Hit-rate@5 cannot answer this question on this corpus

All four modes returned the **same top-5 documents for every query**, so they score identically at 0.917. The Section 4 criterion “hybrid ≥ best single mode” therefore passes on a tie against naïve, not on evidence that hybrid retrieves better.

Ranking differences do appear at **k = 1**. Below that cutoff the four modes are indistinguishable on this corpus, so the honest comparison between them is cost, not quality.

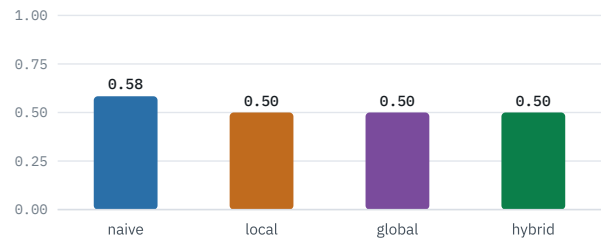
Hit-rate by rank cutoff

Share of queries where a gold document appears in the top k retrieved chunks.



Where ranking does differ

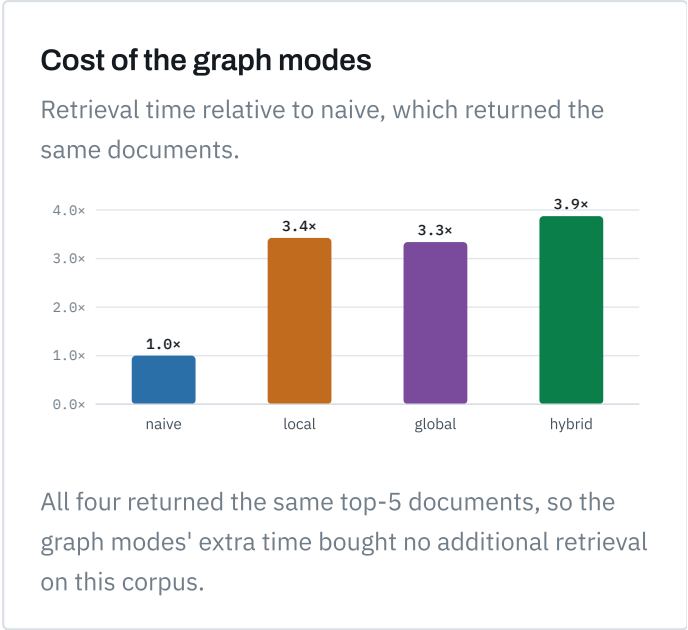
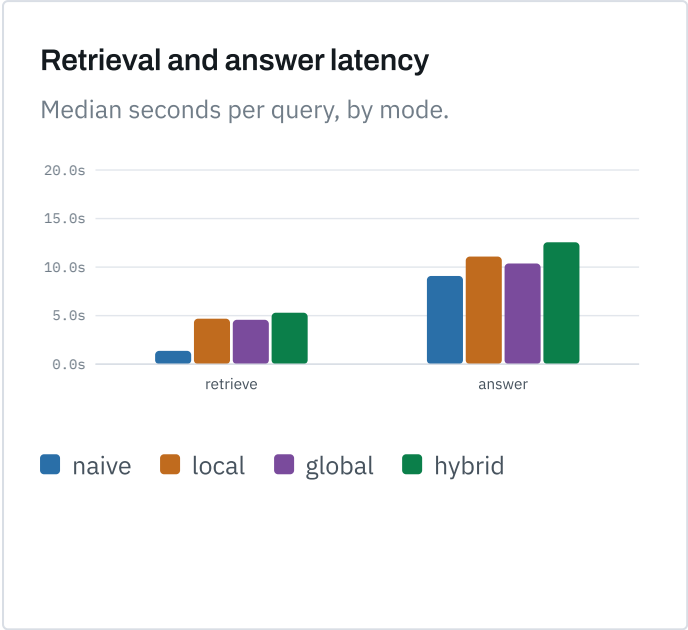
Hit-rate at k = 1, the only cutoff that separates the modes.



By the top-5 cutoff every mode converges on 0.917.

What the modes cost

Identical retrieval, different time to get it — and the cache changes the answer.



Re-run trial 12 queries × 4 modes · LLM cache warm

HIT-RATE@5 · HYBRID 0.917 target ≥ 0.80	HIT-RATE@1 · HYBRID 0.500 naive reaches 0.583	HYBRID RETRIEVAL COST 1.8× vs naive · cache warm	BACKEND UPTIME 100% 32 samples, 0 outages	MODES SEPARATED No identical top-5 across modes
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Do the modes actually differ?

The question Section 4 asks, and the one the headline number answers badly.

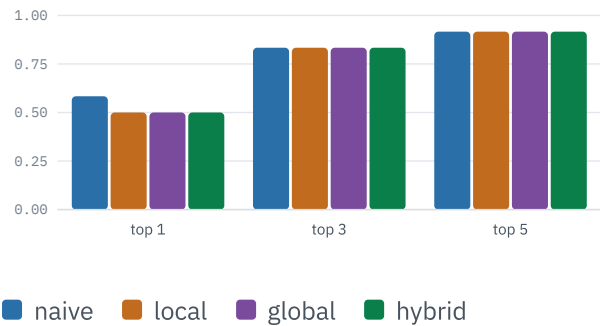
Hit-rate@5 cannot answer this question on this corpus

All four modes returned the **same top-5 documents for every query**, so they score identically at 0.917. The Section 4 criterion “hybrid ≥ best single mode” therefore passes on a tie against naive, not on evidence that hybrid retrieves better.

Ranking differences do appear at **k = 1**. Below that cutoff the four modes are indistinguishable on this corpus, so the honest comparison between them is cost, not quality.

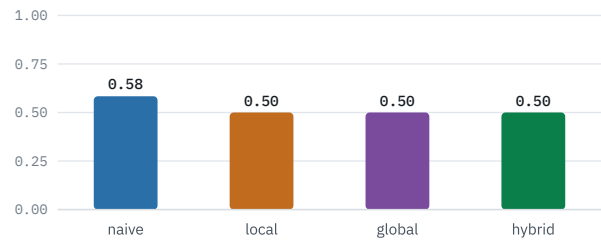
Hit-rate by rank cutoff

Share of queries where a gold document appears in the top k retrieved chunks.



Where ranking does differ

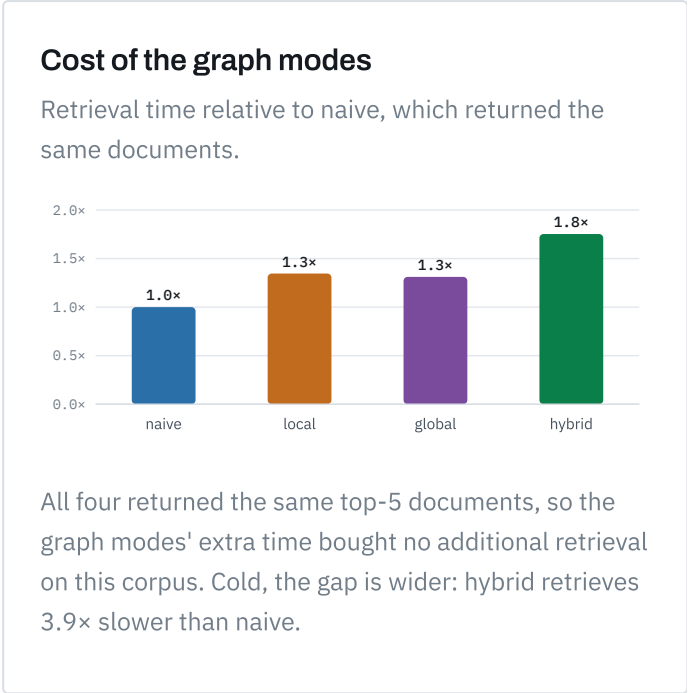
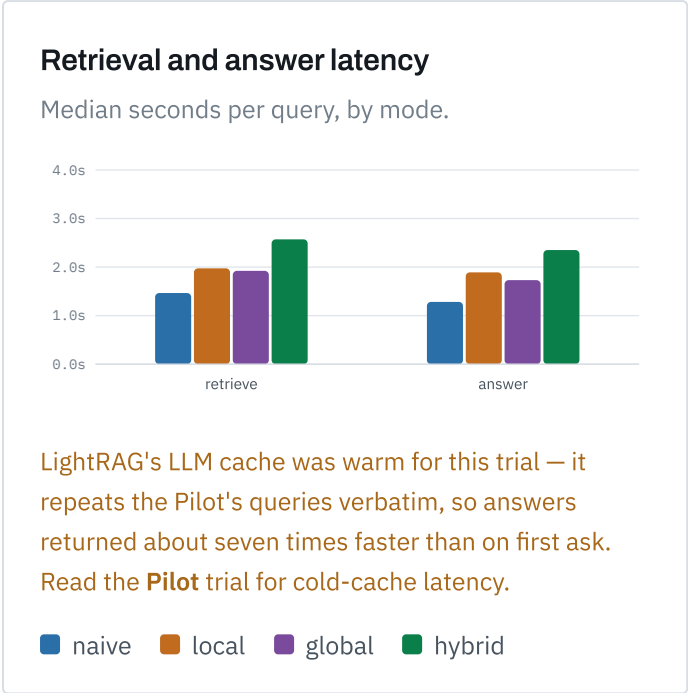
Hit-rate at k = 1, the only cutoff that separates the modes.



By the top-5 cutoff every mode converges on 0.917.

What the modes cost

Identical retrieval, different time to get it — and the cache changes the answer.



Evidence, query by query

Select a row to see what each mode retrieved and answered.

QUERY	ASKS ABOUT	GOLD DOCUMENT	MODES HITTING@5
p01	What line and class length thresholds are suggested as code smell warning signs in .NET?	10-Opinions-For-Creating-More-Ma...	4 of 4
p02	Why is FluentAssertions recommended over the built-in xUnit assertions?	10-Opinions-For-Creating-More-Ma...	4 of 4
p03	Which .NET unit testing framework is advised against, and why?	10-Opinions-For-Creating-More-Ma...	4 of 4
p04	What is the captive dependency problem and what catches it?	10-Opinions-For-Creating-More-Ma...	0 of 4
p05	What does 'build once, deploy many times' imply for branching and configuration?	10-Opinions-For-Creating-More-Ma...	4 of 4
p06	How does parameter binding work for minimal API endpoints, and can it be customised?	Minimal APIs in ASP.NET 9.pdf	4 of 4
p07	What caching technologies are covered for improving minimal API performance?	Minimal APIs in ASP.NET 9.pdf	4 of 4
p08	Which ORM frameworks are demonstrated for data access, and what operations are shown?	Minimal APIs in ASP.NET 9.pdf	4 of 4
p09	How does middleware fit into the request pipeline for minimal APIs?	Minimal APIs in ASP.NET 9.pdf	4 of 4
p10	Is there a relationship between SARS-CoV-2 viral load and how severe a patient's COVID-19 becomes?	Immunity Inflamm Disease - 2021...	4 of 4
p11	How is viral load measured in the COVID-19 studies, and what does the cycle threshold value represent?	Immunity Inflamm Disease - 2021...	4 of 4
p12	Is there anything in this knowledge base about practising handwriting of numbers?	luyen viet so.pdf	4 of 4

Each row expands to the documents every mode retrieved and the answer it produced.

The corpus under test Frozen 2026-09-14. Bar width shows each document's share of the 81 chunks.

<code>Minimal APIs in ASP.NET 9.pdf</code>	61 chunks · 75%
Dominates the corpus: 61 of 81 chunks.	
<code>10-Opinions-For-Creating-More-Maintainable-Dot-Net-Apps.pdf</code>	2 chunks · 2%
Same domain as minimal-apis but only 2 chunks. The hardest target in the corpus and the main discriminator between modes.	
<code>Immunity Inflamm Disease - 2021 - Dadras - The relationship between COVID-19 viral load and disease severity A systematic.pdf</code>	16 chunks · 20%
Different domain; should be trivially separable.	
<code>luyen viet so.pdf</code>	2 chunks · 2%
A Vietnamese handwriting worksheet (digits 0-10 repeated). Almost no factual content, so it serves as a distractor: retrieving it for a substantive question is a false positive.	

What these numbers cannot show

- 1 Every mode retrieved the same top-5 documents for every query, so hit-rate@5 does not discriminate between them here. The “hybrid $\geq$ best single mode” criterion passes on a tie, not on evidence.
- 2 Answer correctness is **not scored on this benchmark**. A blind rating was collected and then withdrawn: it was not the human rating the evaluation plan requires, 82% of its scores sat at the top of the scale, and the only gap between modes came from one query. The generated answers are still here to read; no score is attached to them.
- 3 Twelve queries over four documents is a small sample. Differences at $k = 1$ rest on one or two queries each, so treat the ordering as indicative, not established.
- 4 The corpus was uploaded during development, not curated for evaluation. These numbers characterise the query modes; they are not the reported Section 4 result.
- 5 LightRAG caches LLM results (`enable_llm_cache`), so the Re-run trial — which repeats the Pilot's queries verbatim — answered roughly seven times faster than the Pilot. Cold-cache latency is the Pilot's; any latency target should be set against a cold run.